Experiment No. 5	
SE (CSE)	ROLL NO :
Date of Implementation:	
Aim: Implement multiple linear regression for the given data set	
Programming Language Used : Python	
Upon completion of this experiment, students will be able to CO3 : Perform prediction by applying regression models	

Indicator	Poor	Average	Good
Timeline Maintains submission deadline (2)	Submission not done (0)	One or More than One week late (1)	Maintains deadline (2)
Completion and Organization (2)	Many mistakes in documents (0)	Document is just acceptable (1)	Completed whole document and neatly organized (2)
Program Performance (2)	Could not perform at all (0)	Implemented few parts (1)	Full implementation (2)
Knowledge In depth knowledge of the Experiment (4)	Unable to answer questions(0)	Unable to answer few questions (1-2)	Able to answer all questions with proper explanation (3-4)

Assessment Marks :

Timeline	
Completion and Organization	
Program Performance	
Knowledge	

TOTAL :

EXPERIMENT	***5***
Aim	Implement multiple linear regression for the given data set
Theory	Multiple linear regressions (MLR), is a statistical technique that uses several explanatory variables to predict the outcome of a response variable. The goal of multiple linear regressions is to model the linear relationship between the explanatory (independent) variables and response (dependent) variables. The multiple regression model is based on the following assumptions: • There is a linear relationship between the dependent variables and the independent variables • The independent variables are not too highly correlated with each other • y_i observations are selected independently and randomly from the population • Residuals should be normally distributed with a mean of 0 and variance σ
Implementation	1. Perform the given task in python Consider the following data

X1	10	12	10	9	15	10	14	15	15	16
X2	8	12	9	9	11	9	11	14	11	12
y	10	10	10	9	11	11	9	9	10	10

a) Print first 5 rows of data, Display scatter plot of data
b) Calculate (using formula) and print regression coefficients b0,b1 and b2
c) Display regression line equation
d) Calculate and print coefficient of determination (R squared, Residual sum of squares (RSS), and RMSE
e) Plot regression line
f) Predict the value of y given x1=10, x2=11

2. Perform the given task using python

a) Import the packages numpy, pandas, seaborn and the sklearn.linear_model
b) Read dataset "advertising.csv"
c) Divide data into training and testing split
d) Create an instance of the class LinearRegression, which will represent the regression model
e) Fit the model for training data
f) Get coefficients of regression and coefficient of determination from the model
g) Apply the model for predictions on testing data. Show the residual error plot for training (with green color dots), testing data (blue color) and zero residual error line

Conclusion	
Post Lab Questions:	Write a short note on logistic regression